viva-genesis

Model Upgrade and Implementation Plan

A roadmap for replacing assumption-driven stages with calibrated mechanistic model families, propagating uncertainty, and testing whether a continuous path from geochemistry to cellular organization survives stronger physical and chemical constraints.

Central objective

Move from "this mechanism can pass a gate" to: given empirical constraints and competing mechanistic models, what is the probability, parameter domain, causal mechanism, and experimentally testable prediction for each transition?

Current strength	Main limitation	Upgrade principle
Staged, falsifiable campaign with real simulators and explicit gates.	Several later conclusions follow partly from mechanisms encoded by construction; stages still differ greatly in evidentiary strength.	Use calibrated model ensembles, common biological schemas, uncertainty propagation, and continuous composition across stages.

Source basis. This plan reviews the current investigation-origins-of-life-vent-11 campaign, which now reports 14 run studies spanning vent geochemistry through autonomy, plus bottleneck modules and an adversarial control. The recommendations below distinguish current campaign results from proposed methodological upgrades and external literature/resources.

1. Scientific standard for the next version

The campaign should no longer be optimized around whether each stage can be made to pass. It should be optimized around whether the same conclusion survives stronger chemistry, alternative model structures, uncertainty in poorly known parameters, and propagation of upstream uncertainty into downstream transitions.

For every major transition, report three distinct quantities:

- Feasibility: Does at least one physically admissible parameterization support the transition?
- Robustness: What fraction of the empirically supported parameter distribution supports it, and across which model classes?
- Discrimination: Which observations or perturbations distinguish the favored mechanism from plausible alternatives?

A stage should be considered strong only when its conclusion is not a direct consequence of how the model was encoded. This requires null models, ablations, alternative functional forms, numerical validation, and where possible out-of-sample predictions.

Evidence tiers

Tier	Meaning	Recommended language
Reproduction	Reproduces an external measured or established result	"reproduces / validates"
Mechanistic prediction	Parameterized model makes a new quantitative prediction	"predicts under stated assumptions"
Conceptual demonstration	Reduced model shows that an organizational principle is possible	"demonstrates in a coarse-grained model"

2. Serious model upgrades by campaign stage

Stage	Upgrade	Models / data	Conclusion enabled
0. Vent geochemistry	Thermal-hydrological-chemical reactive transport with evolving mineralogy, porosity and permeability	PFLOTRAN / Reaktoro / PHREEQC; MARHYS 4.0; SUPCRT-style thermodynamic data	Probability that a physically self-consistent vent sustains the required conditions over space and time
I. Carbon fixation	Explicit electrochemistry plus mineral-surface microkinetics	Nernst-Planck; Butler-Volmer; Cantera / CatMAP; 2025-26 Ni/FeS electrochemical data	Whether carbon fixation is limited by thermodynamics, electron delivery, catalytic kinetics or mass transport
II. Protometabolism	Large reaction-network ensembles with reversible kinetics, side reactions and catalyst constraints	Prebiotic reaction-rule networks; libRoadRunner / Cantera; thermodynamic screening	Whether metabolism-like subnetworks are selected by vent chemistry rather than inserted
III. Localization	3-D pore-resolved multiphysics plus reduced pore-network surrogates	FEniCSx / OpenFOAM; PoreSpy / OpenPNM; species-specific Soret data	Whether chemically useful mixtures co-localize in realistic pores
IV. Membranes	Endogenous amphiphile production coupled to assembly, permeability, osmosis and failure	Jordan et al. vesicle phase data; coarse-grained MD for selected permeabilities	Whether the environment can manufacture its own boundary without cutting itself off from resources
IVb. Growth/division	Mechanochemical coupling from reaction flux to osmotic growth and division	Lu et al. autocatalytic compartment data; stochastic partitioning	Whether chemistry itself generates differential reproduction
V. Bioenergetics	Ion-resolved membrane energetics and primitive chemical energy currencies	Poisson/Nernst-Planck; H+/Na+ permeability; thioesters/acyl phosphate/poly-P; Fe redox	When active energy capture becomes advantageous over geological energy supply
VI. Heredity	Stochastic competing inheritance models under matched resource and energy budgets	Compositional inheritance; Gillespie template replication; mutation/error chemistry	When digital heredity outperforms cheaper chemical memory
VII. Ecology	Real chemical networks per protocell exchanging real metabolites	Spatial agents + reaction networks + shared diffusion fields	Whether division of labor and robustness emerge biochemically
VIII. Autonomy	Ablation of geology in the fully integrated persistent world	Continuous PBG world; mechanistic maintenance/resource budgets	Which functions remain necessary across model structures for vent independence

3. Priority upgrade A - a calibrated, evolving vent

Current issue. A Lost City-like window is useful, but the fraction of a chosen parameter grid that passes is not a probability of early-Earth feasibility. Sweep bounds can strongly determine apparent robustness.

Upgrade. Build a hierarchical calibration around observed ultramafic hydrothermal systems. Use MARHYS 4.0 and Lost City measurements to infer distributions for temperature, pH, H2, fluid chemistry, trace metals, fluid/rock ratio and related environmental variables. Fit or constrain uncertain kinetic parameters, then propagate those distributions through reactive transport.

Recommended components

- PFLOTRAN for coupled flow, heat, mineral precipitation/dissolution, aqueous chemistry, sorption, and chemistry-dependent material properties.
- PHREEQC and Reaktoro as independent geochemical implementations; use activity models appropriate to saline hydrothermal fluids and cross-check database dependence.
- Dynamic mineral feedback: precipitation and dissolution change porosity, permeability, tortuosity and reactive surface area.
- Posterior predictive analysis: ask what fraction of vent states consistent with observed systems also satisfy downstream chemical requirements.

Primary output. A spatial-temporal probability field for conditions favorable to downstream chemistry, with uncertainty bars traceable to geochemical parameters and thermodynamic databases.

4. Priority upgrade B - make carbon fixation an electrochemical mechanism

Recent Ni/Fe sulfide vent-analogue experiments provide a route to replace an abstract gradient-efficiency parameter with explicit electrical transport and surface chemistry. The model should resolve fluid redox chemistry -> electrode potential -> overpotential -> current density -> surface reaction rate -> carbon-product flux.

Implement two nested fidelities

- Fast model: Butler-Volmer / Nernst-Planck electrochemistry calibrated to measured current and product yields.
- High-fidelity model: CatMAP- or Cantera-style surface microkinetics with explicit adsorbates, site coverage, catalyst composition and passivation.
- Transport coupling: local H2 and CO2 delivery, ionic strength, temperature and pore-scale diffusion should affect current and reaction rate.

Discovery question. Is usable carbon production controlled primarily by free energy, electronic conductivity, catalyst kinetics, or delivery of reactants? The answer can be tested experimentally by varying Ni/Fe ratio, mineral thickness, current path, pH, H2 and flow.

5. Priority upgrade C - let protometabolism emerge from a larger chemistry

Current issue. A curated rTCA-like network can demonstrate amplification, but cannot show that rTCA-like organization is selected by hydrothermal chemistry.

Upgrade. Construct a prebiotic reaction universe from published reaction transforms and plausible geochemical feedstocks. Annotate reactions with thermodynamics, catalyst requirements, reversible kinetics, side reactions and degradation. Run the network under continuous feed, flow and dilution, then detect persistent flux-carrying and autocatalytic subnetworks.

Required controls

- Randomized networks matched for number of compounds, reactions, energetic span and catalyst density.
- Reaction-graph expansions that include competing sinks and reverse reactions.
- Catalyst-limited and surface-site-limited variants.
- Flux-control analysis to identify rate-controlling reactions and experimentally discriminating perturbations.

Primary conclusion enabled. Whether metabolism-like topology is enriched among the networks that actually persist under vent conditions, rather than being a topology selected in advance.

6. Priority upgrade D - realistic pore physics and multisolute localization

Replace the high-Peclet extrapolation as a headline result with a validated multiscale pore model. Use a 3-D high-fidelity solver for representative pores and a reduced pore-network model for large ensembles. Carry molecule-specific diffusion, thermophoresis, adsorption and degradation.

The scientifically decisive test is not whether one tracer concentrates, but whether the chemically necessary mixture co-localizes. Simulate phosphate, ammonium, formate, acetate, pyruvate, amino acids, amphiphiles and nucleotide precursors simultaneously. Spatial fractionation may be a positive mechanism or a fatal incompatibility.

Numerical requirements

- Mesh-refinement and time-step convergence.
- Head-to-head solver validation inside a defined trustworthy Peclet range.
- Analytic or asymptotic bounds where possible.
- No acceptance gate should depend on a regime outside demonstrated solver convergence.

7. Priority upgrade E - endogenous boundaries and physical reproduction

The membrane stage should no longer begin with amphiphiles already available. Upstream chemistry must generate membrane precursors; transport must concentrate them; assembly occurs only when a composition-dependent threshold is crossed. Vesicle stability and permeability should be parameterized from mixed-amphiphile experiments under alkaline, saline, divalent-cation-rich conditions.

Couple this directly to a physical growth/division model. The Lu et al. autocatalytic-compartment work provides an experimentally grounded bridge: internal autocatalysis can retain products, generate osmotic pressure, drive water influx, produce compartment growth, and convert growth differences into reproductive differences after division.

Target causal chain

reaction flux -> retained solutes -> osmotic imbalance -> water influx -> membrane growth -> instability/division -> differential reproductive success

This eliminates an arbitrary fitness function. Fitness becomes a measurable consequence of chemistry and physics: successful divisions per unit time while maintaining internal organization.

8. Priority upgrade F - primitive energy coupling before modern pumps

Avoid treating a generic "pump" as the first internal bioenergetic innovation. First ask whether simple physicochemical coupling can store environmental free energy and use it to drive unfavorable chemistry.

Candidate mechanisms

- H+ and Na+ gradients across leaky primitive membranes.
- Thioesters and acyl phosphates as chemical energy currencies.
- Polyphosphate / condensed-phosphate chemistry where conditions permit.
- Fe redox coupling and mineral-driven electron-transfer cycles.
- Primitive transport models whose energetic stoichiometry is explicit.

Decision experiment. Compare passive exploitation of an environmental gradient with increasingly active internal coupling. Determine when the energetic benefit of internalization exceeds membrane leakage and maintenance cost.

9. Priority upgrade G - heredity on a common energetic footing

Competing inheritance mechanisms should be evaluated with the same environmental, energetic and material budgets. Compare compositional inheritance, network regeneration, peptide-like inheritance, template replication and hybrids using stochastic kinetics where copy numbers are low.

Use common metrics

- Parent-offspring mutual information in phenotype or functional state.
- Heritable information maintained per unit energy/resource cost.
- Error rate, information capacity and persistence time.
- Population growth under environmental noise and mutation.
- Sensitivity to compartment division and stochastic partitioning.

Discovery question. At what level of chemical complexity does templated, higher-fidelity heredity provide enough additional information capacity to justify its cost?

10. Priority upgrade H - biochemical ecology and autonomy

Ecology. Replace capability vectors with actual reaction networks inside each protocell. Cells exchange specific metabolites through a shared spatial field; cross-feeding, parasitism and division of labor then emerge from flux and permeability rather than symbolic capability transfer.

Autonomy. Treat the current {pump, importer, heredity} result as a hypothesis generator. Re-run vent withdrawal under an ensemble of alternative maintenance functions, transport laws, storage models, permeability assumptions and inheritance mechanisms. The robust result is the intersection of innovations required across model structures, not the minimum set in one reduced budget model.

A stronger autonomy metric

Track the fraction of essential work and material acquisition supplied internally rather than directly by geology. Autonomy is then continuous and can be decomposed by function: energy capture, resource acquisition, catalysis, boundary maintenance, reproduction and information preservation.

11. Cross-cutting computational methods

Method	How to implement	Scientific value
Bayesian calibration	Infer parameter distributions from experiments/field data rather than selecting plausible point values.	Posterior predictive distributions for model observables.
Global sensitivity analysis	Use Morris screening followed by Sobol indices on influential parameters.	Rank controlling mechanisms and experimental priorities.
Structural model ensembles	Run alternative equations/solvers behind common PBG interfaces.	Separate parameter uncertainty from model-form uncertainty.
Uncertainty propagation	Pass distributions and correlated states forward, not just best-fit scalars.	Quantify whether downstream feasibility survives upstream uncertainty.
Numerical verification	Convergence, mass/charge/free-energy residuals, cross-solver comparison.	Prevent numerical artifacts from becoming biological conclusions.
Null and adversarial models	Random networks, alternative environments, mechanism ablations.	Determine whether the claimed mechanism is necessary/specific.
Out-of-sample prediction	Hold back experimental conditions during calibration.	Test whether a model predicts rather than merely reproduces.
Pre-registered gates	Freeze primary metrics and pass/fail logic before major sweeps.	Reduce tuning toward desired outcomes.

Process Bigraph implementation principle

Every upgraded mechanism should implement a canonical biological interface (environment, minerals, molecules, energy, boundary, individual, heredity, ecology). A model class becomes a categorical experimental factor. This lets the same study sweep not only parameter values but also model structure while preserving the surrounding world.

The long-term architecture should preserve one persistent spatial world. Boundary formation, division, extinction and heredity attachment should become actual structural rewrites with conservation ledgers, not stage changes or scalar flags.

12. Build a dedicated viva-genesis data and provenance layer

A curated, machine-readable evidence layer should sit beside the simulation code. Every numerical parameter should carry value or distribution, units, environmental context, source, and provenance class: measured, calibrated, inferred, extrapolated or hypothetical.

High-value datasets and anchors

Resource	Role in viva-genesis
MARHYS 4.0 hydrothermal fluid database	Vent T, pH, H2, major/minor elements, carbon species, isotopes; environmental priors and calibration.
Lost City primary measurements	Ultramafic alkaline vent endmember and local field constraints.
Altair et al. electrochemical vent analogue	Current, catalyst composition, temperature, formate/acetate production; calibrates Stage I.
Panico et al. FeS fuel-cell analogue	Electrochemical coupling across FeS chimney-like interfaces.
Prebiotic reaction-network literature / reaction rules	Candidate chemistry space for emergent network studies.
Baaske / Mast thermal-trap experiments	Pore geometry, Soret behavior, concentration benchmarks.
Jordan et al. mixed amphiphile dataset	Assembly conditions versus pH, salt, divalent ions, temperature and composition.
Lu et al. compartment growth source data	Reaction-driven osmotic growth, competition and reproduction.
Nitrogen / phosphate hydrothermal datasets	Resource availability and activation constraints for C-N-P integration.

13. Implementation roadmap

The roadmap is ordered to increase scientific validity before adding more biological complexity. Each phase should leave behind a reusable Process Bigraph component, a calibrated data package, a numerical-verification notebook, and one or more discriminating studies.

Phase	Build	Implementation	Exit criterion
Phase 1	Environmental calibration	MARHYS-conditioned geochemical ensemble; dynamic reactive transport; database/activity-model comparison	Posterior probability of viable vent states; validated mass/charge/energy ledger
Phase 2	Electrochemical carbon fixation	Butler-Volmer/Nernst-Planck fast model plus selected surface-microkinetic model	Mechanism map: thermodynamic vs electrical vs catalytic vs transport limitation
Phase 3	Open chemical network	Expanded reaction universe, reversible kinetics, side reactions, null networks	Distribution of persistent/autocatalytic subnetworks; motif enrichment vs null
Phase 4	3-D multisolute localization	High-fidelity representative pores plus pore-network surrogate	Co-localization / fractionation phase maps with numerical convergence envelope
Phase 5	Endogenous boundary + growth/division	Amphiphile synthesis -> assembly -> permeability -> osmosis -> reproduction	First continuous transition from chemistry to a growing/dividing individual
Phase 6	Proto-energetics	Primitive energy-currency and ion-coupling models; passive vs active comparison	Energy capture efficiency and threshold for internalization
Phase 7	Competing heredity	Stochastic inheritance models under matched budgets	Information-per-cost frontier; transition to templated heredity
Phase 8	Biochemical ecology	Real network per protocell; metabolite exchange in spatial fields	Emergent cross-feeding, division of labor and robustness
Phase 9	Integrated autonomy	Continuous persistent world; geology withdrawal and model-ensemble ablations	Functions robustly required for independence across model structures

Near-term flagship milestone

Starting from an empirically constrained hydrothermal environment, can inorganic feedstocks produce a persistent chemical reaction network that concentrates, manufactures its own amphiphiles, becomes bounded, grows and divides - with uncertainty propagated from the underlying geochemistry?

14. Concrete software and study deliverables

- vg-data: versioned data registry with machine-readable parameter provenance and distributions.
- vg-geochemical-world: calibrated reactive-transport model with alternative geochemical backends.
- vg-electrochemistry: fast and high-fidelity carbon-fixation models sharing one interface.
- vg-reaction-universe: reaction-rule library, thermodynamic annotations, network generator and null-network tools.
- vg-pore: high-fidelity 3-D reference cases plus reduced pore-network surrogate.
- vg-protoindividual: endogenous amphiphile assembly, permeability, osmotic growth and PBG division rewrite.
- vg-energy: energy ledger and competing primitive coupling mechanisms.
- vg-heredity: common stochastic inheritance interface and information metrics.
- vg-ecology: spatial agents running real chemistry and exchanging real metabolites.
- campaign UQ layer: calibration, Sobol/Morris analysis, structural ensembles, posterior propagation and pre-registered gates.

Decision discipline

- Do not move a headline conclusion forward if it depends on a numerically unvalidated regime.
- Do not call a mechanism "necessary" until it survives alternative model structures.
- Do not call a pathway "emergent" when its topology or capability was inserted by design.
- Treat incompatibility between stages as a primary discovery, not a failure of the campaign.
- Prioritize experiments that discriminate between mechanisms over experiments that simply tighten one fitted parameter.

15. Scientific endpoint

The strongest eventual result is not that all stages pass. It is that a continuous, uncertainty-bounded path exists - or fails to exist - from geochemical disequilibrium to a reproducing and heritable compartment across multiple mechanistic model classes.

A defensible high-level conclusion would be:

Across experimentally constrained parameter distributions and multiple mechanistic model classes, viva-genesis identifies where geological disequilibrium can support carbon fixation, self-amplifying chemistry, localization, boundary formation, growth, primitive inheritance and ecological persistence - and identifies the specific bottlenecks, incompatibilities and environmental features that break that path.

Selected references and resources

These sources motivate the proposed upgrades; they are not evidence that the full geochemistry-to-cell sequence occurred historically.

MARHYS 4.0 hydrothermal fluid database. PANGAEA dataset; DOI landing page for large hydrothermal fluid compilation.

PFLOTRAN Theory Guide. Reactive transport with mineral precipitation/dissolution, sorption, aqueous kinetics, and chemistry-dependent material properties.

Altair et al. (2025). Carbon Reduction Powered by Natural Electrochemical Gradients - vent-like Ni/Fe sulfide electrochemistry and carbon products.

Panico et al. (2026). A short-circuited fuel cell reproduces hydrothermal FeS chimney conditions for the onset of life. Communications Chemistry.

Wołos et al. (2020). Synthetic connectivity, emergence, and self-regeneration in the network of prebiotic chemistry. Science.

Baaske et al. (2007). Extreme accumulation of nucleotides in simulated hydrothermal pore systems. PNAS 104:9346-9351. DOI 10.1073/pnas.0609592104.

Jordan et al. (2019). Promotion of protocell self-assembly from mixed amphiphiles at the origin of life. Nature Ecology & Evolution 3:1705-1714. DOI 10.1038/s41559-019-1015-y.

Lu et al. (2024). Small-molecule autocatalysis drives compartment growth, competition and reproduction. Nature Chemistry 16:70-78. DOI 10.1038/s41557-023-01276-0.

Report note. Current-campaign statements are based on the uploaded investigation-origins-of-life-vent-11.html. External methods and datasets listed here are proposed upgrades and literature resources, not components already implemented in the current campaign.